Applied Bayesian Statistics in Animal & Biological Sciences

1) Introduction to Bayesian reasoning

Henk J. van Lingen

hvanling@uoguelph.ca

May 11, 2026

Introduction to myself

My positions now and previously

- PhD on (dynamic) modeling of ruminal methane production and anaerobic microbial physiology
- I also worked as a postdoc in Computational Systems Biology
- I'm currenty a postdoc at the Centre for Nutrition Modelling with Jen Ellis

UNIVERSITY OF GUELPH

Introduction to myself

My positions now and previously

- PhD on (dynamic) modeling of ruminal methane production and anaerobic microbial physiology
- I also worked as a postdoc in Computational Systems Biology
- I'm currenty a postdoc at the Centre for Nutrition Modelling with Jen Ellis

I have expertise in Bayesian Statistics (in STAN and MCMCglmm)

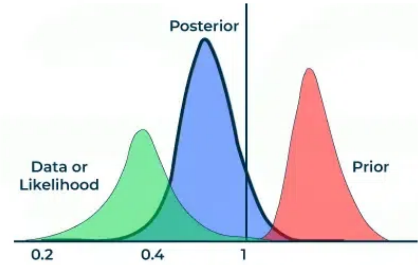
- I'm an experienced biological systems modeler, I'm not a statistician
- Disclaimer: I may not be able to exhaustively explain all concepts used in this course that I took from papers or books

Introduction of the 21 course participants

- Are you a master student, PhD candidate, postdoc or faculty member?
- What's your main research topic?
- Did you use Bayesian statistics before? In WinBUGS, MCMCglmm, BGLR, brms, STAN or PROC MCMC?
- Do you run your analyses in R, Python, C++, Julia or SAS?
- What reasons did you have to sign up for this course? Any specific analysis in mind?

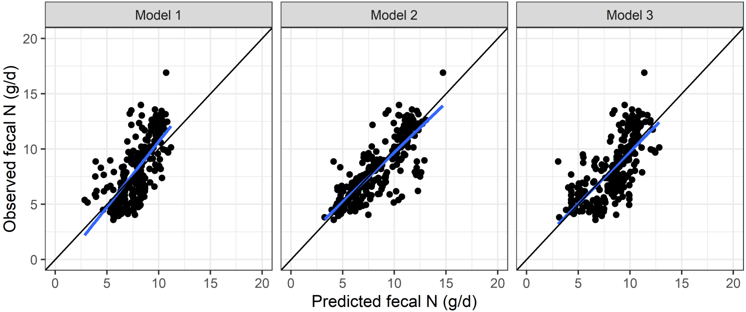
Today's program

- Bayesian statistics examples in Animal science
- Frequentist statistics vs Bayesian statistics
- Bayesian reasoning
- The Bayes' rule



UNIVERSITY OF GUELPH

Application in Sheep Nutrition



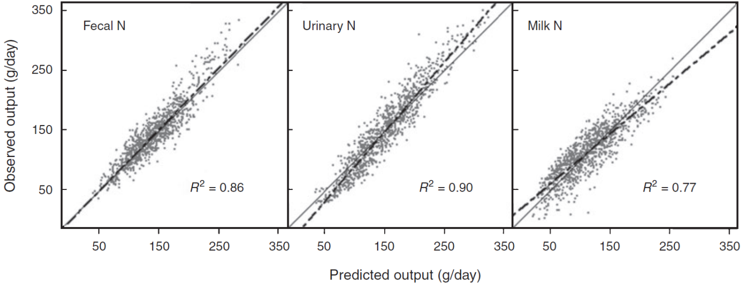
$$y_{\text{ij}} = x_{\text{ij}}^T \beta + s_i + e_{\text{ij}} \text{ with } s_i \sim N(0, \sigma_s^2), e_{\text{ij}} \sim N(0, \sigma_e^2) \tag{1}$$

Model	RMSPE
1] -1.16±0.90 + 5.72±0.64 × DMI + 0.196±0.026 × CP	24.5%
2] -4.36±0.90 + 6.03±0.48 × DMI + 0.109±0.021 × CP + 0.115±0.008 × NDF	23.7%
3] 12.3±1.0 + 5.88±0.43 × DMI - 0.166±0.009 × DMD + 0.123±0.018 × CP	19.3%

Van Lingen et al., 2021

UNIVERSITY OF GUELPH

Application in Dairy Nutrition - Predicting N excretion



$$Y_{\mathrm{ijk}} \sim N_3(\mu_{\mathrm{ijk}}, \Sigma) \tag{2}$$

Y_{ijk} contains N excretion for k records, j animals and i studies, μ_{ijk} a function of a matrix of covariates X_{ijk}, Σ residual (co)-variance matrix

Reed et al., 2014

UNIVERSITY OF GUELPH

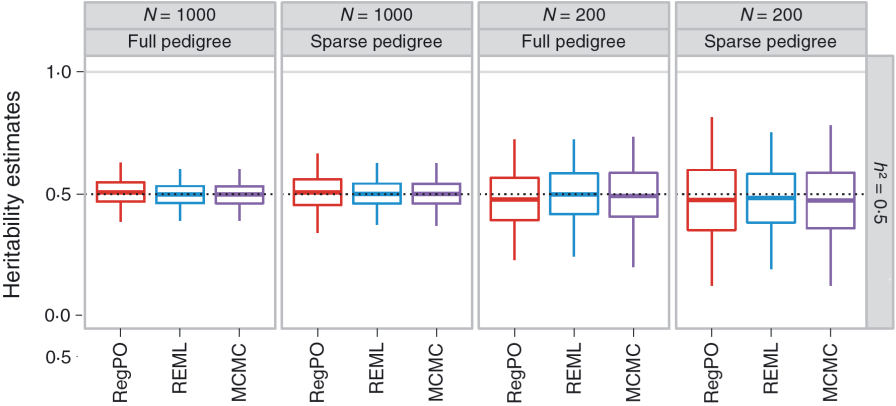
Application in Dairy Nutrition - Multivariate N prediction

$$\Sigma_1 = \begin{bmatrix} \sigma^2_{\mathrm{FN}} & 0 & 0 \\ 0 & \sigma^2_{\mathrm{UN}} & 0 \\ 0 & 0 & \sigma^2_{\mathrm{MN}} \end{bmatrix}$$

$$\Sigma_2 = \begin{bmatrix} \sigma^2_{\mathrm{FN}} & \sigma_{\mathrm{FN}}\sigma_{\mathrm{UN}} & \sigma_{\mathrm{FN}}\sigma_{\mathrm{MN}} \\ \sigma_{\mathrm{FN}}\sigma_{\mathrm{UN}} & \sigma^2_{\mathrm{UN}} & \sigma_{\mathrm{UN}}\sigma_{\mathrm{MN}} \\ \sigma_{\mathrm{FN}}\sigma_{\mathrm{MN}} & \sigma_{\mathrm{UN}}\sigma_{\mathrm{MN}} & \sigma^2_{\mathrm{MN}} \end{bmatrix}$$

- The non-zero off-diagonal case Σ_2 quantifies residual variance correlation, whereas the zero off-diagonal case Σ_1 does not

Application in Breeding & Genetics - Heritability



- Linear mixed model to assess the genetic variance component
- Similar frequentist (RegPO,REML) and Bayesian (MCMC) inference?

De Villemereuil et al., 2013

UNIVERSITY OF GUELPH

Frequentist P-values as a measure of evidence

P-value is not the probability H_0 is false ...

- Probability of observing the result or more extreme result under H_0
- Then a P-value is also based on "non-observed" data

UNIVERSITY OF GUELPH

Frequentist P-values as a measure of evidence

P-value is not the probability H_0 is false ...

- Probability of observing the result or more extreme result under H_0
- Then a P-value is also based on "non-observed" data
- Example: CH_4 from cattle fed control and inhibitor diets:
  - $H_0 : \Delta = \mu_{\mathrm{CH_4,control}} - \mu_{\mathrm{CH_4,inhibitor}} = 0$
  - $\hat{\Delta} = 60$ g/d may yield $t_{\mathrm{obs}} = 2.19$ with a P-value for the probability $t_{\mathrm{obs}} \geq 2.19$, given that H_0 is true

UNIVERSITY OF GUELPH

Frequentist P-values as a measure of evidence

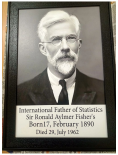
P-value is not the probability H_0 is false ...

- Probability of observing the result or more extreme result under H_0
- Then a P-value is also based on "non-observed" data
- Example: CH_4 from cattle fed control and inhibitor diets:
  - $H_0 : \Delta = \mu_{\mathrm{CH_4,control}} - \mu_{\mathrm{CH_4,inhibitor}} = 0$
  - $\hat{\Delta} = 60$ g/d may yield $t_{\mathrm{obs}} = 2.19$ with a P-value for the probability $t_{\mathrm{obs}} \geq 2.19$, given that H_0 is true

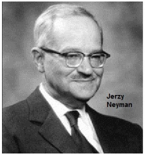
P-values are not effect sizes ... :(

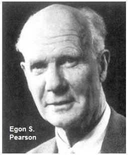
Frequentist paradigms: 95% confidence interval instead?

- Fisher: strength of evidence to reject H_0 based on a P-value
- Neyman-Pearson: H_0 vs H_1 decision, minimize Type I (false positive) and Type II (false negative) errors



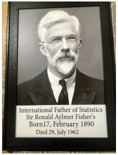
vs



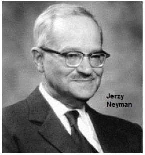


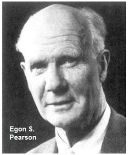
Frequentist paradigms: 95% confidence interval instead?

- Fisher: strength of evidence to reject H_0 based on a P-value
- Neyman-Pearson: H_0 vs H_1 decision, minimize Type I (false positive) and Type II (false negative) errors



vs





Instead, consider the Bayesian view on probability:

- There is a 95% chance that the true effect size is in the *credible interval* (more intuitive)
- One can even accompany the credible interval with the full distribution of the true effect size

Idea of Bayesian reasoning in daily life

Visiting Belgium for the first time:

- You may hardly know anything about Belgians (no prior knowledge)

Idea of Bayesian reasoning in daily life

Visiting Belgium for the first time:

- You may hardly know anything about Belgians (no prior knowledge)
- Travel books tell you Belgians produce excellent beers and chocolate (prior knowledge)

UNIVERSITY OF GUELPH

Idea of Bayesian reasoning in daily life

Visiting Belgium for the first time:

- You may hardly know anything about Belgians (no prior knowledge)
- Travel books tell you Belgians produce excellent beers and chocolate (prior knowledge)
- You meet Belgians during your visit (data)

Idea of Bayesian reasoning in daily life

Visiting Belgium for the first time:

- You may hardly know anything about Belgians (no prior knowledge)
- Travel books tell you Belgians produce excellent beers and chocolate (prior knowledge)
- You meet Belgians during your visit (data)
- You learn the Belgian cuisine provides various delicious savoury dishes and you update your opinion (posterior knowledge)

UNIVERSITY OF GUELPH

Basic idea of the Bayesian approach: trial example

Check the efficacy of a new mouthwash:

- First result: 25% plaque reduction [10%, 40%] compared to tap water

Basic idea of the Bayesian approach: trial example

Check the efficacy of a new mouthwash:

- First result: 25% plaque reduction [10%, 40%] compared to tap water
- Result seems great, but previous products resulted in reductions of only 5% to 15%

Basic idea of the Bayesian approach: trial example

Check the efficacy of a new mouthwash:

- First result: 25% plaque reduction [10%, 40%] compared to tap water
- Result seems great, but previous products resulted in reductions of only 5% to 15%
- Use this past experience as prior knowledge and combine with the (new) data

Basic idea of the Bayesian approach: trial example

Check the efficacy of a new mouthwash:

- First result: 25% plaque reduction [10%, 40%] compared to tap water
- Result seems great, but previous products resulted in reductions of only 5% to 15%
- Use this past experience as prior knowledge and combine with the (new) data
- Update your knowledge on the efficacy of the new mouthwash

Bayesian introduction to the 21 course participants

- Who has used:
  - Mixed models

Bayesian introduction to the 21 course participants

- Who has used:
  - Mixed models
  - Non-linear regression

Bayesian introduction to the 21 course participants

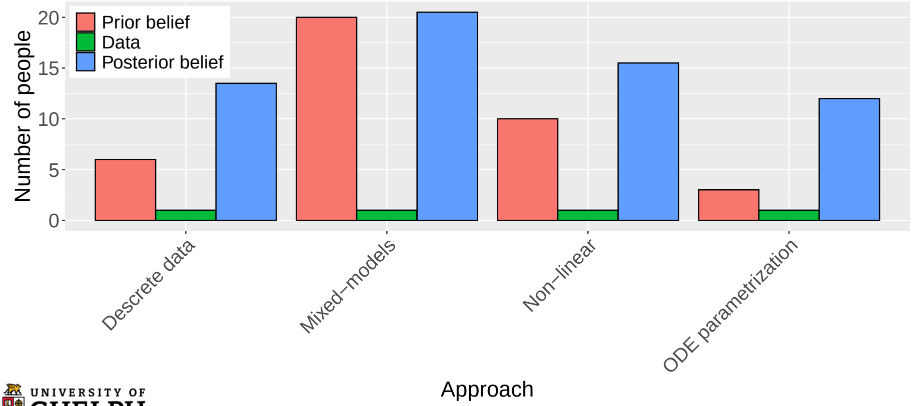
- Who has used:
  - Mixed models
  - Non-linear regression
  - Discrete data models

Bayesian introduction to the 21 course participants

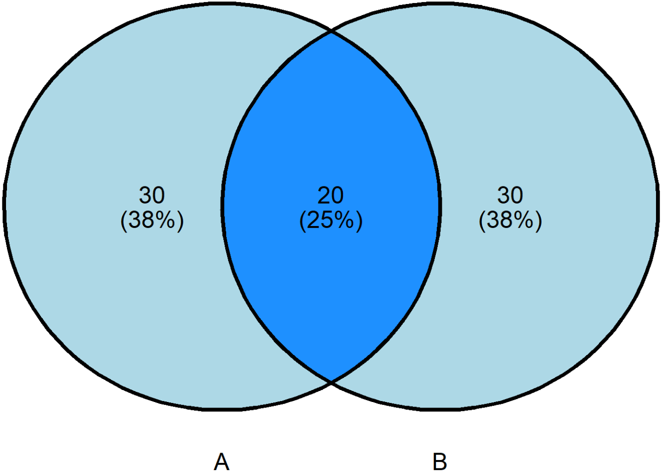
- Who has used:
  - Mixed models
  - Non-linear regression
  - Discrete data models
  - Parameter estimation of an ODE model?

Bayesian introduction to the 21 course participants

- Who has used:
  - Mixed models
  - Non-linear regression
  - Discrete data models
  - Parameter estimation of an ODE model?



Events A and B and their (joint) elementary probabilities



- Suppose: Event A is a positive diagnostic, B a diabetic patient
- Intersection or joint probability of A and B: $P(A, B)$

UNIVERSITY OF GUELPH

Probability theory and Bayes' theorem or Bayes' rule

Elementary property in probability theory:

- Marginal probabilities: $P(A)$ and $P(B)$; conditional probabilities: $P(A|B)$ and $P(B|A)$
  - $P(A|B) = \dfrac{P(A,B)}{P(B)}$ and $P(B|A) = \dfrac{P(A,B)}{P(A)}$
  - $P(A,B) = P(A) \cdot P(B|A) = P(B) \cdot P(A|B)$

Probability theory and Bayes' theorem or Bayes' rule

Elementary property in probability theory:

- Marginal probabilities: $P(A)$ and $P(B)$; conditional probabilities: $P(A|B)$ and $P(B|A)$
  - $P(A|B) = \dfrac{P(A,B)}{P(B)}$ and $P(B|A) = \dfrac{P(A,B)}{P(A)}$
  - $P(A,B) = P(A) \cdot P(B|A) = P(B) \cdot P(A|B)$

Bayes' rule follows from elementary property:

- $P(B|A) = \dfrac{P(A|B) \cdot P(B)}{P(A)}$
  - Note: $P(B|A)$ quantifies the probability that a patient is diabetic given a positive diagnostic

Verify/apply the Bayes' rule using Boston city hospital data

Test	Diabetic	Non-diabetic	Total
+	56	49	105
-	14	461	475
Total	70	510	580

- $$P(B|A) = \frac{P(A|B) \cdot P(B)}{P(A)} = \frac{\frac{56}{70} \cdot \frac{70}{580}}{\frac{105}{580}} = \frac{\frac{56}{580}}{\frac{105}{580}} = \frac{56}{105}$$

UNIVERSITY OF GUELPH

Verify/apply the Bayes' rule using Boston city hospital data

Test	Diabetic	Non-diabetic	Total
+	56	49	105
-	14	461	475
Total	70	510	580

- $$P(B|A) = \frac{P(A|B)\cdot P(B)}{P(A)} = \frac{\frac{56}{70}\cdot\frac{70}{580}}{\frac{105}{580}} = \frac{\frac{56}{580}}{\frac{105}{580}} = \frac{56}{105}$$

Law of total probability: $P(A) = P(A|B)\cdot P(B) + P(A|B^C)\cdot P(B^C)$

UNIVERSITY OF GUELPH

Verify/apply the Bayes' rule using Boston city hospital data

Test	Diabetic	Non-diabetic	Total
+	56	49	105
-	14	461	475
Total	70	510	580

- $P(B|A) = \dfrac{P(A|B) \cdot P(B)}{P(A)} = \dfrac{\frac{56}{70} \cdot \frac{70}{580}}{\frac{105}{580}} = \dfrac{\frac{56}{580}}{\frac{105}{580}} = \dfrac{56}{105}$

Law of total probability: $P(A) = P(A|B) \cdot P(B) + P(A|B^C) \cdot P(B^C)$

- Yields: $P(B|A) = \dfrac{P(A|B) \cdot P(B)}{P(A|B) \cdot P(B) + P(A|B^C) \cdot P(B^C)}$

UNIVERSITY OF GUELPH

Verify/apply the Bayes' rule using Boston city hospital data

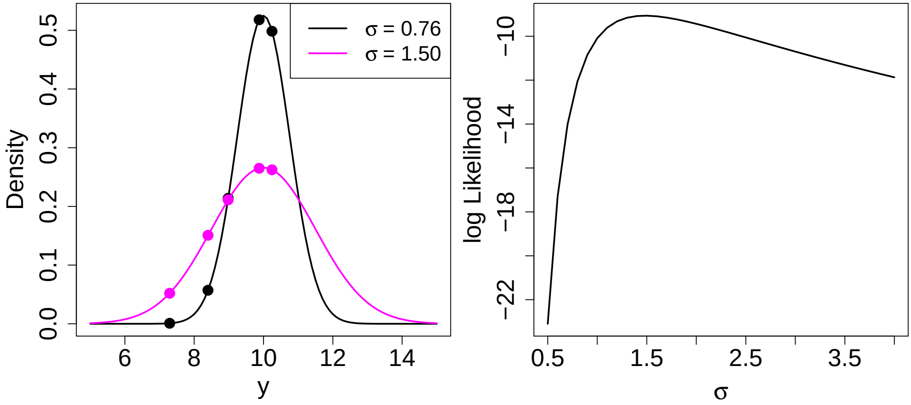
Test	Diabetic	Non-diabetic	Total
+	56	49	105
-	14	461	475
Total	70	510	580

- $$P(B|A) = \frac{P(A|B) \cdot P(B)}{P(A)} = \frac{\frac{56}{70} \cdot \frac{70}{580}}{\frac{105}{580}} = \frac{\frac{56}{580}}{\frac{105}{580}} = \frac{56}{105}$$

Law of total probability: $P(A) = P(A|B) \cdot P(B) + P(A|B^C) \cdot P(B^C)$

- Yields: $$P(B|A) = \frac{P(A|B) \cdot P(B)}{P(A|B) \cdot P(B) + P(A|B^C) \cdot P(B^C)}$$
- If $P(B) = 3\%$ worldwide, $$P(B|A) = \frac{\frac{56}{70} \cdot 0.03}{\frac{56}{70} \cdot 0.03 + \frac{49}{461} \cdot 0.97} = 0.20$$

Likelihood is used in Frequentist and Bayesian inference



- (log) Likelihood is the plausibility of model parameter values given the observed data: $L(\boldsymbol{\theta}|\boldsymbol{y}) = P(\boldsymbol{y}|\boldsymbol{\theta}) = \Pi_{i=1}^{n}\, p(y_i|\boldsymbol{\theta})$
    - $l(\mu = 10; \sigma = 1.50|\boldsymbol{y}) = \log(0.052 \times 0.15 \times 0.21 \times 0.27 \times 0.26) = -9.1$
    - $l(\mu = 10; \sigma = 0.76|\boldsymbol{y}) = -12.9$

UNIVERSITY OF GUELPH

Model fitting: apply Bayes' theorem

For parameters that follow continuous distributions:

$$\overbrace{p(\boldsymbol{\theta}|\boldsymbol{y}) = \frac{p(\boldsymbol{\theta})L(\boldsymbol{\theta}|\boldsymbol{y})}{p(\boldsymbol{y})}}^{\text{Bayes' rule}} = \frac{p(\boldsymbol{\theta})L(\boldsymbol{\theta}|\boldsymbol{y})}{\int p(\boldsymbol{\theta})L(\boldsymbol{\theta}|\boldsymbol{y})d\boldsymbol{\theta}} \propto p(\boldsymbol{\theta})L(\boldsymbol{\theta}|\boldsymbol{y}) \tag{3}$$

Model fitting: apply Bayes' theorem

For parameters that follow continuous distributions:

$$\overbrace{p(\boldsymbol{\theta}|\boldsymbol{y}) = \frac{p(\boldsymbol{\theta})L(\boldsymbol{\theta}|\boldsymbol{y})}{p(\boldsymbol{y})}}^{\text{Bayes‘ rule}} = \frac{p(\boldsymbol{\theta})L(\boldsymbol{\theta}|\boldsymbol{y})}{\int p(\boldsymbol{\theta})L(\boldsymbol{\theta}|\boldsymbol{y})d\boldsymbol{\theta}} \propto p(\boldsymbol{\theta})L(\boldsymbol{\theta}|\boldsymbol{y}) \tag{3}$$

This simplifies to:

$$\underbrace{p(\boldsymbol{\theta}|\boldsymbol{y})}_{\text{posterior}} \propto \underbrace{p(\boldsymbol{y})L(\boldsymbol{\theta}|\boldsymbol{y})}_{\text{prior times likelihood}} \tag{4}$$

Model fitting: apply Bayes' theorem

For parameters that follow continuous distributions:

$$\overbrace{p(\boldsymbol{\theta}|\boldsymbol{y}) = \frac{p(\boldsymbol{\theta})L(\boldsymbol{\theta}|\boldsymbol{y})}{p(\boldsymbol{y})}}^{\text{Bayes' rule}} = \frac{p(\boldsymbol{\theta})L(\boldsymbol{\theta}|\boldsymbol{y})}{\int p(\boldsymbol{\theta})L(\boldsymbol{\theta}|\boldsymbol{y})d\boldsymbol{\theta}} \propto p(\boldsymbol{\theta})L(\boldsymbol{\theta}|\boldsymbol{y}) \tag{3}$$

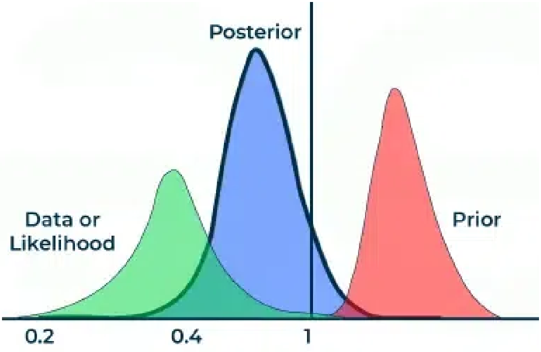
This simplifies to:

$$\underbrace{p(\boldsymbol{\theta}|\boldsymbol{y})}_{\text{posterior}} \propto \underbrace{p(\boldsymbol{y})L(\boldsymbol{\theta}|\boldsymbol{y})}_{\text{prior times likelihood}} \tag{4}$$

- Parameters are considered random variables that may follow any distribution
- Optimizing $p(\boldsymbol{\theta})L(\boldsymbol{\theta}|\boldsymbol{y})$, i.e. data and prior knowledge, generates knowledge
- Powerful stochastic Markov-chain Monte Carlo algorithms generate the posterior

UNIVERSITY OF GUELPH

Prior, likelihood and posterior



- Prior distribution: what you believed before seeing data
- Likelihood: what the data say is plausible/new evidence
- Posterior distribution: the updated belief after combining both

Reasons for choosing Bayesian statistics

1. Full Probability Statements Over Outcomes/Parameters - Not just one point value
2. Prior usage is natural
3. Prediction & Decision Making
4. MCMC is a powerful optimization procedure

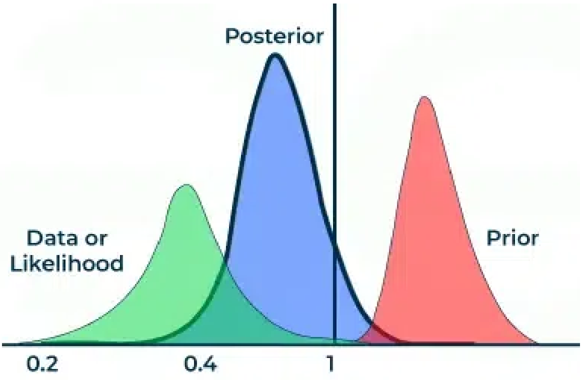
Reasons for choosing Bayesian statistics

1. Full Probability Statements Over Outcomes/Parameters - Not just one point value
2. Prior usage is natural
3. Prediction & Decision Making
4. MCMC is a powerful optimization procedure

- Bayesian statistics is computationally demanding
- Bayesian statistics requires more knowledge about probability distributions

UNIVERSITY OF GUELPH

Thank you for your attention!



Questions?